GEN4 at Jarama: lap time and energy projection from a Gen3 Evo lap simulation calibrated on the 2026 Madrid E-Prix

Carlos Sánchez Maceda 9 November 2026 `github.com/cjsanchezmaceda/FE_Energy_LapSim`

Summary. A quasi-steady-state (QSS) lap simulation of the Formula E layout at Jarama was calibrated on the best dry FP1 lap of the 2026 Madrid E-Prix and used to project the GEN4 car ahead of the 16–20 November pre-season test.

- **Gen3 Evo calibration:** one free parameter (tyre $\mu = 1.689$) reproduces the 350 kW lap (89.403 s); the independent 300 kW lap is predicted to within +0.14 s. The result is insensitive to the unknown drag area and mass.
- **GEN4, unconstrained: 80.6 s** in qualifying trim (600 kW, high-DF), −8.8 s vs Gen3 Evo; high-DF is 1.7 s faster than low-DF at 600 kW.
- **GEN4, energy-limited (450 kW):** the model supports the intended use of the two kits (high-DF for qualifying, low-DF for races): **low-DF is faster for any race budget below ≈3.6 kWh/lap**, i.e. any race longer than 14 laps on the 51.25 kWh usable energy; by 3.5 s at 2.6 kWh/lap. Lift-and-coast meets the same budget 0.9–3.6 s/lap faster than capping drive power (2.4 s for Gen3 Evo).

1 Model

Track. Track limits digitised from satellite imagery (Google Earth, ±0.6 m vs GPS boundaries), including the FE loop (+30.1 m). Centreline 3933 m vs 3934 m official. Racing line: minimum-curvature path (Heilmeier et al., 2020) with 1.3 m margin to the limits, 3882 m, minimum radius 11.6 m; curvature from a periodic cubic spline, 2 m resolution.

Lap simulation. Point-mass QSS: cornering limit $v^2 = \mu g/(|\kappa| - \mu\rho C_L A/2m)$, backward (braking) and forward (traction) passes on a two-lap domain. Combined grip via the friction circle; longitudinal load transfer (Fh/L) in traction and braking; battery-side power limits with drive/regen efficiencies; Gen3 rear axle braked by regen only (350 kW), GEN4 with rear friction brakes. Energy: $E = \sum P\,\Delta t$ from the wheel power, net of regeneration.

Table 1: Vehicle parameters. *Public*: FIA regulations / Formula E. *Assumed*: not public, quoted with the range tested.

	Gen3 Evo	GEN4	Source
Mass incl. driver [kg]	862	1046	FIA Tech. Regs (public)
Wheelbase / width [m]	2.97 / 1.70	3.08 / 1.79	FIA Tech. Regs (public)
Usable race energy [kWh]	38.5 + 3.85 PIT BOOST	51.25	public
Drive power [kW]	300 race / 350	450 race / 600	public
Driven axles	RWD; AWD at 350 kW	AWD	public
Regen [kW] front / rear	250 / 350	350 / 350	FIA Tech. Regs (public)
Rear friction brakes	no	yes	public
Tyre μ	1.689 (calibrated)	1.689 × 1.05	tyre factor assumed, 1.00–1.10
$C_D A$ [m^2] low / high-DF	0.90	0.95 / 1.24	assumed; high-DF ≈+30 % drag
$C_L A$ [m^2] low / high-DF	1.6	2.7 / 3.8	assumed
Front motor [kW]	50	250	FIA Tech. Regs (public)
η drive / regen	0.93 / 0.90	0.93 / 0.90	assumed

2 Gen3 Evo calibration (Madrid E-Prix 2026, FP1, dry)

Only the tyre friction coefficient is fitted. Target: best 350 kW lap of FP1 (Ticktum, 1:29.403, best of 16 drivers who ran 350 kW). Check: best 300 kW lap (Da Costa, 1:31.612, best of only 4 drivers, hence an upper bound rather than a limit lap).

Table 2: Calibration and robustness. Each row re-fits μ on the 350 kW lap.

Case	μ	300 kW lap [s]	vs FIA 91.612 s
Nominal ($C_D A$ 0.90, AWD at 350 kW, 862 kg)	**1.689**	**91.76**	**+0.14**
$C_D A$ 0.75 / 1.05	1.675 / 1.703	91.70 / 91.83	+0.08 / +0.22
Mass 840 / 936 kg	1.674 / 1.737	91.77 / 91.73	+0.16 / +0.12
350 kW in FP1 without front motor (RWD)	1.734	90.83	−0.78

$C_D A$ and mass are absorbed by μ with ≤ 0.07 s effect on the prediction. The main open question is whether the front motor was active on the 350 kW FP1 laps. If it was (nominal), the model reproduces the FIA gap between the 350 and 300 kW laps (2.36 vs 2.21 s). If it was not, μ rises to 1.734 and the model would place the best 300 kW lap 0.8 s slower than the car's limit, which is less plausible.

3 GEN4 projection, unconstrained lap

Table 3: GEN4 flat-out laps, FE layout. Δ vs Gen3 Evo 350 kW (89.40 s).

Kit	Power [kW]	Lap [s]	Δ [s]	v_{max} [km/h]	v_{min} [km/h]	Energy [kWh]
low-DF	450	84.97	−4.42	273	52	3.61
low-DF	600	82.26	−7.14	293	52	4.36
high-DF	450	83.65	−5.74	267	52	4.37
high-DF	**600**	**80.59**	**−8.80**	290	52	5.21

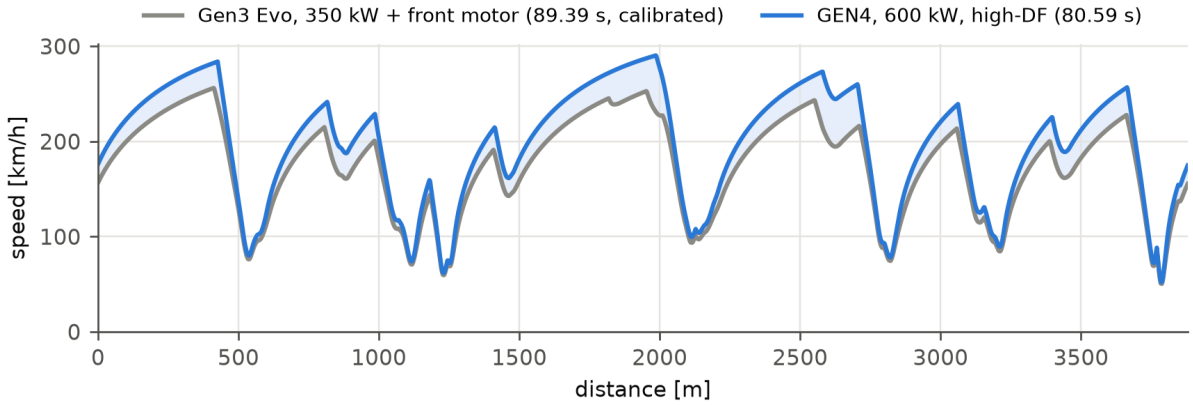


Figure 1: Speed vs distance: Gen3 Evo 350 kW (calibrated) vs GEN4 600 kW high-DF. Minimum speeds in the slow corners are almost unchanged (52 vs 50 km/h): the GEN4 gains come from power on the straights (+25–40 km/h at the end of each, top speed 290 vs 256 km/h) and from downforce in the medium-speed corners.

4 Energy-limited laps

Two ways of meeting an energy budget per lap. *Lift-and-coast*: the driver lifts off over the last fraction f_c of each acceleration phase, measured in distance back from the flat-out braking point (no drive, no regen). *Power cap*: lower maximum drive power over the whole lap. f_c or the cap is solved so the lap uses exactly the budget.

Gen3 Evo check vs the race. Budget $(38.5 + 3.85)/23 = 1.84$ kWh/lap (usable energy + PIT BOOST, 23 laps).

Table 4: Gen3 Evo, 300 kW, Madrid race energy budget.

Strategy	Lap [s]	Energy [kWh]	Setting
Flat-out (no budget)	91.75	2.34	–
Lift-and-coast	**95.18**	1.84	$f_c = 43\,\%$
Power cap	97.55	1.84	186 kW
Race: best normal laps, 4 of 5 drivers	94.8–95.7	–	excl. lap 1, pit, ATTACK MODE
Race: median normal lap	98.2	–	80 laps, 5 drivers

The energy-limited lift-and-coast lap (95.18 s) falls within the best normal race laps of four of the five drivers transcribed, and 3.0 s below the median, consistent with traffic and racing effects that the model does not include. Power capping would cost 2.4 s/lap more.

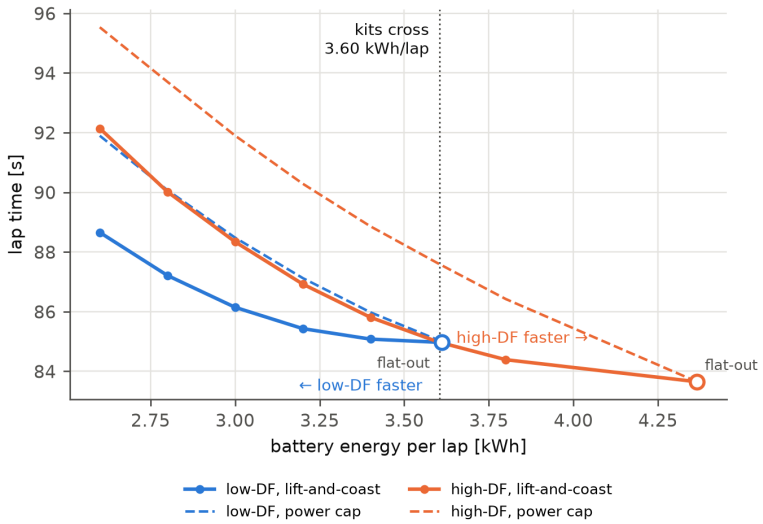


Figure 2: GEN4 at 450 kW: lap time vs battery energy per lap, by aero kit and saving strategy. The kits cross at 3.60 kWh/lap: below it, low-DF is faster. On the 51.25 kWh GEN4 usable energy, 3.6 kWh/lap corresponds to a 14-lap race.

Table 5: GEN4 at 450 kW, lap time [s] for a given energy per lap. Low-DF needs no saving above its flat-out 3.61 kWh.

Energy [kWh/lap]	2.6	3.0	3.4	3.8	flat-out
low-DF	88.66	86.15	85.08	84.97	84.97
high-DF	92.14	88.34	85.81	84.39	83.65
Δ (high − low)	+3.48	+2.19	+0.73	−0.58	−1.32
Power cap, low-DF	91.90	88.48	85.98	84.97	84.97
Power cap, high-DF	95.53	91.90	88.86	86.42	83.65

5 Does the kit conclusion survive the assumptions?

Table 6: Δ lap time high-DF − low-DF [s], one assumption changed at a time. Negative: high-DF faster.

Case	flat-out, 450 kW	3.0 kWh/lap, lift-and-coast
Nominal	−1.32	+2.20
Tyre factor 1.00 / 1.10	−1.38 / −1.27	+1.80 / +2.09
High-DF C_LA −15 / +15 %	−0.36 / −2.21	+2.34 / +1.77
High-DF C_DA −10 / +10 %	−1.59 / −1.06	+0.92 / +3.52

The sign of the conclusion holds in every case: high-DF is faster on an unconstrained lap, low-DF is faster once energy is limited. The high-DF drag area is the assumption that matters most for the race case (0.9–3.5 s); downforce matters most for the flat-out lap. The GEN4 test speed traps would pin C_DA down directly.

6 Limitations and next steps

Limitations. Point-mass QSS (no transients, no tyre temperature); single μ for both axles; front-motor traction uses static front load; lift-and-coast is a fixed fraction per acceleration phase, not an optimal energy allocation; drivetrain efficiencies and all GEN4 aero and tyre values are assumed; GEN4 regen split assumed; whether 350 kW FP1 running used the front motor is unconfirmed; track digitised, not surveyed.

Next steps. (1) Re-fit tyre factor and C_DA from GEN4 test laps and speed traps; the model then becomes a GEN4 tool. (2) Optimal-control energy allocation (where to coast, lap by lap) instead of a uniform fraction. (3) ATTACK MODE and PIT BOOST deployment strategy over a race distance. (4) Slipstream: energy saved per lap when following.